

Airspan Automation & Tools Development Intern - Detailed Learning Roadmap

Python Programming

- Variables and Data Types
- Operators and Expressions
- Conditional Statements
- Loops
- Functions
- Strings
- Lists
- Tuples and Sets
- Dictionaries
- OOP: Classes and Objects
- OOP: Constructors
- OOP: Inheritance
- OOP: Polymorphism
- OOP: Encapsulation
- File Handling
- Exception Handling
- Modules and Packages
- datetime module
- os module
- sys module
- subprocess module
- JSON
- Regular Expressions
- Requests library
- API calls
- Automation Scripts

Linux / Bash

- pwd, ls, cd
- mkdir, rmdir
- cp, mv, rm
- cat, head, tail

- grep
- find
- chmod
- Processes (ps, top, kill)
- Environment variables
- Shell scripting basics
- Loops and conditions in Bash
- Executing Python scripts from terminal

Git & GitHub

- Version Control Concepts
- git init
- git clone
- git status
- git add
- git commit
- git push
- git pull
- git branch
- git checkout
- git merge
- Resolving merge conflicts
- GitHub repositories
- README creation
- Project hosting

Robot Framework

- Introduction
- Architecture
- Test Cases
- Keywords
- Variables
- Test Suites
- Executing Tests
- Reports

APIs & Automation

- HTTP Basics

- GET Requests
- POST Requests
- Status Codes
- Headers
- JSON Responses
- Python Requests Library
- Reading Log Files
- Generating Reports
- Automation Projects

CI/CD Basics

- What is CI/CD
- Build Pipeline
- Testing Pipeline
- Deployment Pipeline
- GitHub Actions
- Jenkins Basics

Project Work (Student Resource Hub)

- Upload to GitHub
- Professional README
- Screenshots
- Document APIs
- JWT Authentication explanation
- MongoDB explanation
- CRUD explanation
- Interview-ready project explanation